

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 20 August 2026, 8:45 AM**State** Finished**Completed on** Thursday, 20 August 2026, 8:47 AM**Time taken** 1 min 22 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int main() {
3      int n, count = 0;
4      if (scanf("%d", &n) != 1) return 0;
5      int c = 0; count++;
6      for (int i = n / 2; i < n; i++) {
7          count++;
8          for (int j = 1; j < n; j = 2 * j) {
9              count++;
10             for (int k = 1; k < n; k = k * 2) {
11                 count++;
12                 c++;
13                 count++;
14             }
15             count++;
16         }
17         count++;
18     }
19     count++;
20     printf("%d", count);
21     return 0;
22 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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